

MBI6013 RESEARCH PROPOSAL

**Donor- and cohort-resolved single-cell analysis
of compositional and transcriptional B-cell
remodeling in systemic lupus erythematosus**

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Outcome-locked, sample-level and reproducible by design

Project overview

Systemic lupus erythematosus (SLE) is associated with altered B-cell differentiation, interferon activation and expansion of antibody-secreting and atypical B-cell compartments. Single-cell studies can resolve these processes, but two biologically different phenomena are often conflated: changes in the relative abundance of B-cell states and changes in gene expression within the same state. In addition, cell-level inference, outcome-informed annotation and imbalanced technical cohorts can produce apparently strong but poorly generalizable disease signals.

This project will reanalyse a large public SLE single-cell dataset from audited raw counts using a donor-, sample-, library- and cohort-resolved design. B-cell states will be reconstructed and annotated without access to disease outcomes. After those states are frozen, SLE-associated composition and within-state transcription will be estimated separately at the biological-sample level. Discovery definitions will then be applied without refitting to an independent SLE dataset. The project is designed to produce an honest hierarchy of robust, cohort-specific and inconclusive findings rather than to force all evidence into a single pathogenic-cell narrative.

1. Background and significance

SLE is a heterogeneous autoimmune disease in which loss of B-cell tolerance, autoantibody production, innate immune sensing and type I interferon signalling interact across patients and disease stages. Peripheral B-cell abnormalities include changes in naive and memory compartments, expansion of plasmablasts and antibody-secreting cells, and enrichment of CD11c-high, T-bet-positive or double-negative phenotypes [1-4]. These populations are biologically relevant, but their nomenclature and boundaries vary across studies [5]. A transcriptional cluster should therefore not be labelled pathogenic merely because it is enriched in cases or resembles a literature signature.

Perez et al. profiled more than 1.2 million peripheral blood mononuclear cells from SLE cases and controls and demonstrated cell-type-specific disease and genetic associations [1]. The resource is unusually valuable for B-cell analysis because it includes many donors, samples and multiplexed libraries. The same complexity also creates a statistical challenge: technical processing cohorts are not uniformly balanced for disease, some donors contribute repeated samples, and biological samples are distributed across multiple libraries. Pooled cell-level tests or a single cohort-adjusted disease coefficient can therefore confound disease with design structure.

Single-cell disease analysis also requires a sharp distinction between composition and cell-intrinsic state. An increased number of cells expressing an interferon program may result from expansion of an interferon-high state, upregulation of interferon genes within several stable states, or both. Subpopulation-specific pseudobulk methods were developed precisely because cells from the same individual are not independent replicates [6,7]. Likewise, cell-state abundance is compositional: an

increase in one state necessarily changes the relative abundance of others, motivating sample-level models and compositional sensitivity analyses [8].

Mechanistic studies nominate plausible regulators of atypical or age-associated B-cell programs. ZEB2 is required for age-associated B-cell differentiation, and TLR7-FTO-linked metabolic regulation can promote this fate [2,3]. These studies provide valuable biological priors. However, overlap with a ZEB2 or TBX21 signature in observational single-cell data does not establish causal regulation. The proposed work will therefore treat pathway and regulon evidence as hypothesis prioritization and reserve causal language for perturbational or strong causal-genetic evidence.

The significance of the project lies in its inferential separation of three questions: which B-cell states can be reconstructed without disease knowledge; which of those states change in abundance within cohorts that support a direct case-control contrast; and which states change transcription internally. This framework can clarify apparently conflicting B-cell findings and can remain informative even when a visually distinctive state fails external replication.

2. Central hypothesis and specific aims

Central hypothesis

SLE remodels peripheral B cells through separable changes in the abundance of neutral B-cell states and in transcription within those states. The two layers will show different magnitudes and different degrees of external reproducibility.

Aim 1. Reconstruct stable, disease-blind B-cell states from audited raw counts

Rationale. Outcome-informed labels and inherited whole-PBMC embeddings can make disease enrichment circular. A de novo B-cell representation is required before disease comparisons.

Approach. Hard-QC-passing B cells will be extracted from the authoritative raw counts, with a full-PBMC marker audit checking extraction completeness. Residual doublet risk will be scored per complete library without automatic re-deletion. PCA, unintegrated and Harmony-adjusted graphs, UMAP and Leiden clustering will be rebuilt within B cells. Neutral states will be frozen using markers, biological/technical coverage, bridge-sample concordance and resampling; an ISG-excluded reconstruction will test identity stability while disease fields remain physically separated.

Expected outcome. A reproducible neutral B-cell reference with explicit state uncertainty, technical-contaminant flags and a frozen mapping object for external data.

Aim 2. Separate SLE-associated composition from within-state transcription

Rationale. State abundance and state-internal expression represent distinct biological processes and require different sample-level models.

Approach. State counts will be aggregated by biological sample for cohort-resolved compositional models. Raw counts will be aggregated by sample and frozen state for pseudobulk SLE-control differential expression. The main contrast will be restricted to strict single-processing-cohort samples in the processing cohort with adequate case-control support; a second supported cohort will be analysed separately as exploratory. A global composition test will precede state-specific effects. Pathway and regulon analyses will be applied to ranked within-state disease effects, not to state-marker contrasts.

Expected outcome. A state-by-process map that distinguishes composition-dominant, transcription-dominant, concordant and unsupported B-cell signals in SLE.

Aim 3. Test frozen discovery states and signatures in independent SLE data

Rationale. Recomputing labels or thresholds inside validation data can convert validation into a second discovery analysis.

Approach. A discovery-only classifier or reference-mapping model will be evaluated by donor-stratified cross-validation and frozen with an uncertainty rule. It will then be applied to GSE135779 without refitting to disease labels. Mapped state abundance and prespecified within-state/signature effects will be tested separately in childhood and adult strata before any combined estimate. GSE163121 will provide directional evidence only because of its small sample size, and OneK1K will be used only as healthy reference context.

Expected outcome. A transparent replication scorecard separating replicated, directionally consistent, inconclusive and non-replicated findings.

3. Preliminary data and feasibility

No original wet-lab data are claimed. All results below arise from local audits and disease-blind feasibility analyses of public data.

3.1 Authoritative discovery object

The B-cell source object contains 152,981 cells and 30,172 genes. The audited `raw/X` matrix is non-negative and integer-valued. Metadata identify 259 donors, 271 biological samples, 88 technical libraries and four processing cohorts. Eleven donors have repeated samples, all samples are multiplexed across multiple libraries, and 53 samples bridge processing cohorts. These observations define the sample as the biological unit, the donor as a correlation unit and the library as a technical unit.

3.2 Common-support audit

Disease is strongly imbalanced across processing cohorts. In the ambiguity-free subset of donors represented by exactly one biological sample and one processing cohort ($n = 195$), normal/SLE counts are 28/0, 1/87, 5/8 and 41/25 in cohorts 1-4, respectively. Cohort 4 is the primary direct comparison,

cohort 3 is exploratory, and cohorts 1-2 are discovery-only strata. This programmatically reproduced audit prevents a misleading pooled disease effect.

3.3 Full raw-count preparation

Frozen hard-QC rules retained 150,402 cells and all 30,172 genes in a new disease-blind full object. The object contains only source cell index, donor, sample, library and processing cohort. Disease, disease state and `ct_cov` are stored separately. A read-back audit confirmed integer raw counts and no protected-field leakage.

3.4 Disease-blind smoke reconstruction

A 20,000-cell smoke analysis recovered coherent naive-like, memory-like, atypical-like and plasmablast-like marker structure. Harmony reduced local same-library, same-sample and same-processing-cohort neighbor fractions. At a diagnostic Leiden resolution of 0.6, all seven provisional clusters were represented across many donors and libraries. This supports feasibility of a full representation rebuild.

The smoke run also exposed an important failure mode: Scrublet was applied after balanced sampling and predicted an implausibly high median doublet fraction of 14.8% per library. These calls have been rejected for freezing. The corrected workflow now runs Scrublet on complete libraries before any sampling or graph construction. This negative result demonstrates that the gate system is preventing an analysis artifact from entering the manuscript.

4. Research design and methods

4.1 Data resources

Role	Resource	Intended use	Interpretive boundary
Discovery	Perez et al., GSE174188 [1]	state reconstruction, composition and within-state DE	disease contrasts restricted by cohort support
Main validation	GSE135779 [9]	frozen state mapping and sample-level replication	no threshold refitting in validation
Directional validation	GSE163121	direction and marker context	two controls and three SLE samples; not confirmatory
Healthy reference	OneK1K [10]	B-cell maturation and population context	not SLE replication
Mechanistic context	ZEB2, FTO/TLR7 and EBV/APC studies [2,3,11]	candidate-regulator interpretation	observational overlap is not causality

4.2 Source integrity and quality control

The source path, SHA256 checksum, matrix dimensions, matrix encoding and metadata dictionaries will be recorded before analysis. Hard-QC thresholds are frozen at at least 500 total counts, at least 200 detected genes, no more than 10% mitochondrial counts, no more than 1% haemoglobin counts, no more than 0.5% platelet-marker counts, and detection of at least one B-lineage marker. Every cell receives a reason-level QC record; no source file is overwritten.

These thresholds are intentionally conservative because the source object is an already extracted B-cell compartment. A lightweight full-PBMC B-marker audit will test whether a material B-like population was excluded by source labels. Distributional plots and state-specific retention will be examined to ensure that plasmablasts or other high-RNA states are not preferentially removed. The all-hard-QC branch will remain available for sensitivity analysis.

4.3 Complete-library doublet assessment

Because the source workflow already applied donor demultiplexing and doublet handling, Scrublet will be rerun per complete library only as a residual-risk diagnostic with a prespecified expected rate and deterministic seed. Cell scores, automatic thresholds, predicted rates and failures will be retained. Automatic calls will not be applied until rate distributions and mixed-lineage marker enrichment are reviewed. Analyses will compare approved singlets with all hard-QC cells. A library with an extreme rate will trigger diagnostic review rather than an arbitrary rate cap.

4.4 Disease-blind representation and batch assessment

Library-aware recurrent highly variable gene selection will be performed on normalized log expression while preserving raw counts for pseudobulk. PCA and a neighbor graph will first be generated without integration. Harmony will then adjust the PCA representation using technical library. Both representations will be retained.

Integration quality will be evaluated with local neighbor mixing for library, sample and processing cohort, bridge-sample consistency and conservation of known B-cell marker structure. Harmony will not be used to infer disease effects or to erase processing-cohort restrictions. A representation will pass only if technical mixing improves without collapsing biologically coherent rare states. An otherwise matched reconstruction excluding strong interferon-stimulated genes will test whether state identity is stable rather than merely an IFN activation gradient.

4.5 State definition and freeze

Leiden clustering will be evaluated across a prespecified resolution grid. Candidate states must satisfy four criteria: coherent canonical markers, representation across biological samples and donors, absence of dominance by a single library, and assignment stability under cell resampling or graph perturbation.

Platelet-, erythroid- or mixed-lineage-enriched clusters will be flagged as technical/contaminant states rather than forced into B-cell biology.

Neutral names such as *naive*, *memory*, *atypical-like* and *plasmablast* may be assigned only after marker review. Disease-enriched names are prohibited. State IDs, names, marker sets, excluded states and the external mapping model will be versioned and frozen before protected outcomes are joined.

4.6 Cohort-resolved composition

For each biological sample, the number of cells in every frozen state and the total number of eligible B cells will be calculated. The primary per-state analysis will use a beta-binomial mixed model of state versus other-B-cell counts within processing cohort 4, with disease and prespecified covariates and a donor random intercept for repeated samples where identifiable. A global sample-level compositional test will precede state-specific interpretation.

Bridge samples will be reserved for technical concordance and excluded from the primary disease coefficient. Processing cohort 3 will be analysed separately as exploratory. Cohorts 1 and 2 will not contribute a direct disease coefficient. Sensitivities will retain one sample per donor, vary minimum cell requirements, compare approved singlets with all hard-QC cells, and use a centered-log-ratio or Bayesian compositional method such as scCODA [8]. Effects, intervals and false-discovery-rate-adjusted P values will be reported; cells will never be treated as independent disease replicates.

4.7 Sample-by-state pseudobulk differential expression

Raw counts will be summed by biological sample and frozen state after technical library contributions from the same sample are combined. Sample-state strata must meet a prespecified minimum cell count. Lowly expressed genes will be filtered using pseudobulk expression, followed by TMM normalization and a voom-dream or equivalent model that accounts for donor correlation. The primary cohort 4 disease coefficient and the exploratory cohort 3 coefficient will be estimated separately.

This contrast tests SLE versus control within a fixed state. State-versus-state marker tests will be retained only for annotation. Multiple testing families will be declared before results are interpreted. Sample influence, mean-variance fit and residual diagnostics will be inspected.

4.8 Pathway and regulatory interpretation

Gene-set enrichment will use ranked within-state disease effects and curated Hallmark, Gene Ontology and immune pathways. Regulon activity may be estimated with a curated resource such as DoRothEA/decoupleR. Candidate regulators will be ranked by convergence across within-state DE, regulon activity, external direction and published mechanistic evidence. This ranking will generate testable hypotheses; it will not be described as genetic or experimental validation.

4.9 Frozen external validation

A state mapper will be trained exclusively in discovery data using intersecting genes. Donor-stratified cross-validation will quantify class discrimination, calibration and uncertainty. Low-confidence cells will be assigned **unmapped** rather than forced into a discovery state. All coefficients, marker signatures, directions and thresholds will be frozen before external outcomes are examined.

In GSE135779, mapping quality will first be evaluated without disease comparison. Childhood and adult strata will then be estimated separately for sample-level state abundance and within-state or signature effects using the same direction convention as discovery, followed by compatibility assessment or a prespecified meta-effect where justified. A finding will be termed replicated only if mapping quality is acceptable, the direction was prespecified and the external sample-level estimate supports that direction. Otherwise it will be classified as directional, inconclusive or not replicated.

4.10 Reproducibility and outcome lock

The workflow uses timestamped run directories, immutable source objects, checksums, deterministic seeds, environment files and tidy Source Data tables. Analysis proceeds through four freezes: cells/representation, neutral states, discovery outcomes and external validation. Protected outcomes are joined only after the neutral-state freeze. Existing v6 outputs remain archived for provenance and will not be overwritten or manually altered to match v7.

5. Expected results and interpretation

The expected primary contribution is not necessarily the discovery of a new B-cell subtype. Rather, the study should reveal whether prominent SLE B-cell signals are driven by abundance, within-state activation or both. A robust interferon-associated transcriptional signal may reproduce even if an atypical-like abundance effect is cohort specific. Conversely, a stable composition change may occur without a large within-state transcriptional effect. Either outcome would clarify the level at which SLE remodels B cells.

The highest-impact outcome would combine a stable neutral state, a supported cohort 4 composition or within-state effect, a directionally consistent cohort 3 estimate and frozen external replication in GSE135779. A negative external result will still be informative if the mapper is well calibrated and the discovery design is rigorous, because it will delimit generalizability and prevent a cohort-specific observation from being promoted as universal.

6. Potential difficulties and alternatives

Extreme doublet rates. Automatic thresholds may fail in individual libraries. Scores and mixed-lineage enrichment will be reviewed, and all-hard-QC and approved-singlet branches will be compared. Rates will not be clipped to a desired value.

Over-integration. Harmony may remove real biology. Unintegrated and Harmony representations will be compared, bridge samples will provide technical diagnostics, and final states must retain marker coherence in both views.

Rare states. A rare cluster may lack enough sample-state strata for pseudobulk. It may be merged only when marker and stability evidence support a broader state; otherwise it will remain descriptive and excluded from formal disease DE.

Residual cohort confounding. No global pooled disease coefficient will be used. Primary and exploratory cohorts will be reported separately, and cohorts without both groups will not estimate disease effects.

Repeated donors and incomplete covariates. Mixed models and one-sample-per-donor sensitivity will address repeated samples. Covariates will be included only when measured consistently and supported within the analysis cohort; they will not be imputed merely to obtain a more complex model.

A paired repeated-donor analysis may follow the core freeze but will not be interpreted as treatment-causal because time, activity and therapy may co-vary.

External mapping failure. Low-confidence cells will be allowed to abstain. If exact state transfer is not supported, validation will be restricted to frozen gene-set activity and clearly described as program-level rather than state-level replication.

No mechanistic validation. Public observational data cannot establish causality. The final study will prioritize a short regulator list for future perturbation in primary B cells or appropriate models.

7. Milestones and decision gates

Phase	Deliverable	Pass criterion
Gate C2B1	complete-library doublet diagnostics	interpretable rate/score distributions and approved cell policy
Gate C2B2	full disease-blind representation	improved technical mixing with conserved B-cell biology
Gate C2C	frozen neutral states	marker coherence, broad coverage and resampling stability
Gate C3	composition and pseudobulk outcomes	valid cohort-specific sample-level models and diagnostics
Gate C4	independent validation	frozen mapping quality and prespecified replication scorecard
Manuscript freeze	five main figures, Source Data and v7 text	every claim linked to a frozen table and analysis unit

8. Expected impact

The proposed work will provide a reusable framework for separating cellular composition from state-internal disease biology in complex public single-cell datasets. For SLE, it will clarify which B-cell observations are robust across samples and cohorts, which are mainly transcriptional, and which remain context-dependent. The resulting claim hierarchy should be suitable for a high-quality genomics or computational-biology journal and will define a focused set of hypotheses for future functional validation.

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